DEP512E – HW1 Part A
Newmark–β Analysis of a SDOF System

Mesut ADALI - 801262107

19.10.2026

1 Method

The equation of motion is

$$m\ddot{x} + c\dot{x} + kx = p(t), \qquad p(t) = -m\ddot{x}_g(t) \text{ for ground motion.}$$

The Newmark–β method (incremental form) is used with $\beta = 1/4$ and $\gamma = 1/2$. At each time step:

$$\hat{k} = k + \frac{\gamma}{\beta\Delta t}c + \frac{m}{\beta\Delta t^2}, \qquad \Delta\hat{p}_i = \Delta p_i + A\,\dot{x}_i + B\,\ddot{x}_i, \qquad \Delta x_i = \frac{\Delta\hat{p}_i}{\hat{k}}$$

$$A = \frac{m}{\beta\Delta t} + \frac{\gamma}{\beta}c, \qquad B = \frac{m}{2\beta} + \Delta t\left(\frac{\gamma}{2\beta} - 1\right)c$$

$\Delta\dot{x}_i$ and $\Delta\ddot{x}_i$ follow from Δx_i, and $x_{i+1} = x_i + \Delta x_i$ (similarly for $\dot{x}$ and $\ddot{x}$).

2 Code

- `newmark(m,c,k,p,dt,beta,gamma,x0,v0)`: applies the steps above and returns the displacement, velocity and acceleration vectors.
- Main part: defines the system and the loads, calls `newmark` and plots the results. The spring force is $f_S = kx$ and the damper force is $f_D = c\dot{x}$.
- System: $m = 1$ kg, $\omega_n = 2\pi$ rad/s, $\xi = 0.05$, so $k = m\omega_n^2 = 39.48$ N/m and $c = 2\xi m\omega_n = 0.628$ N s/m.
- Problem 1: $p(t) = 2\sin(\pi t/2)$, $\Delta t = 0.01$ s, 20 s.
- Problem 2: ground acceleration from `ground_acc.txt`, $\Delta t = 0.005$ s (7860 points, 39.3 s), $p = -m\ddot{x}_g$.

3 Results – Problem 1

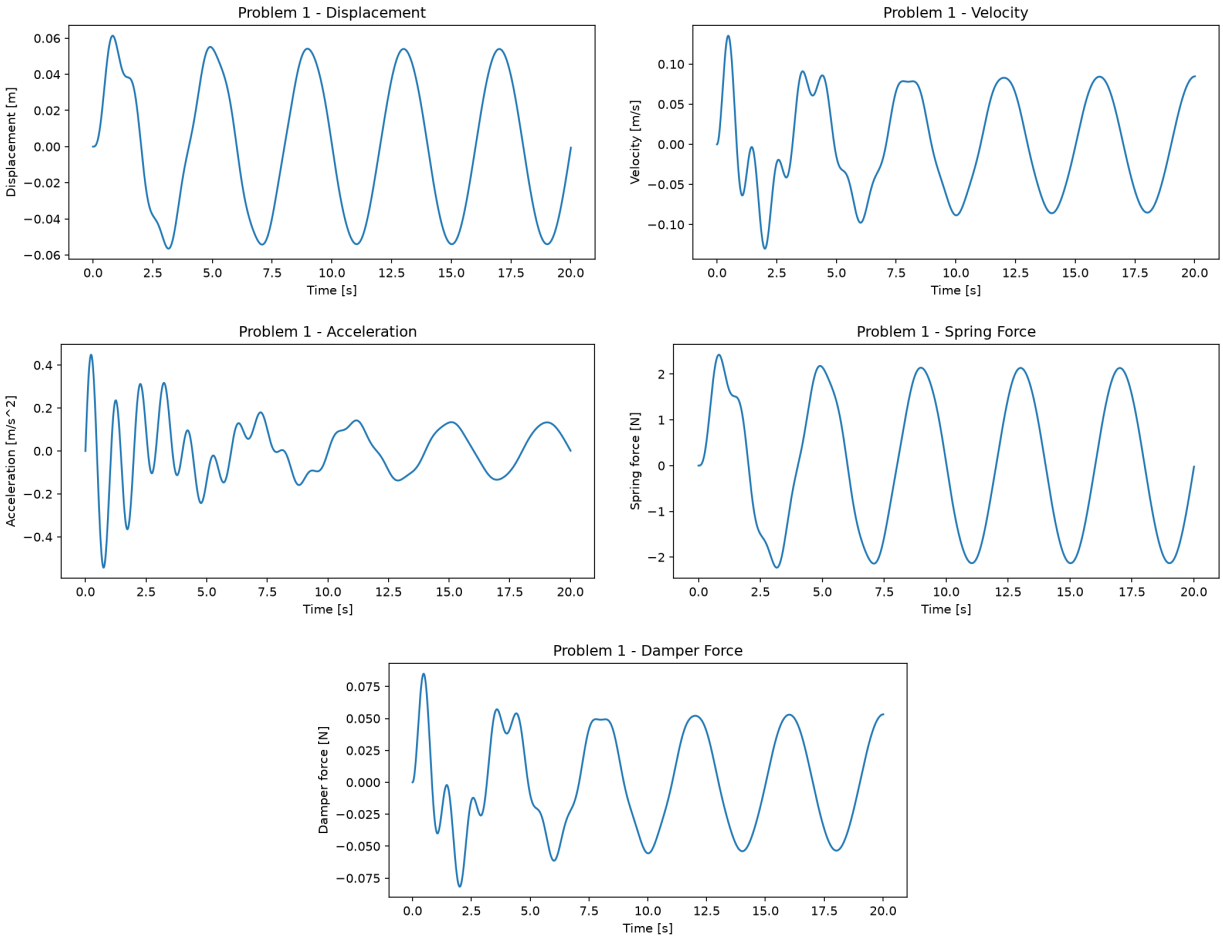


Figure 1: Problem 1: time histories.

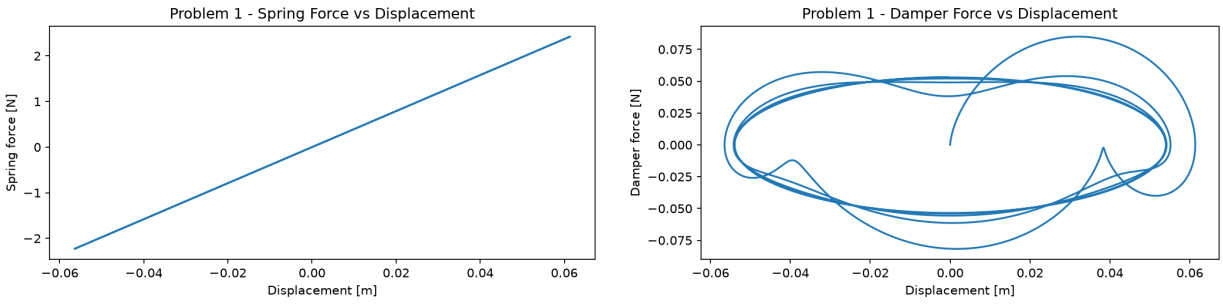


Figure 2: Problem 1: displacement vs spring force and damper force.

4 Results – Problem 2

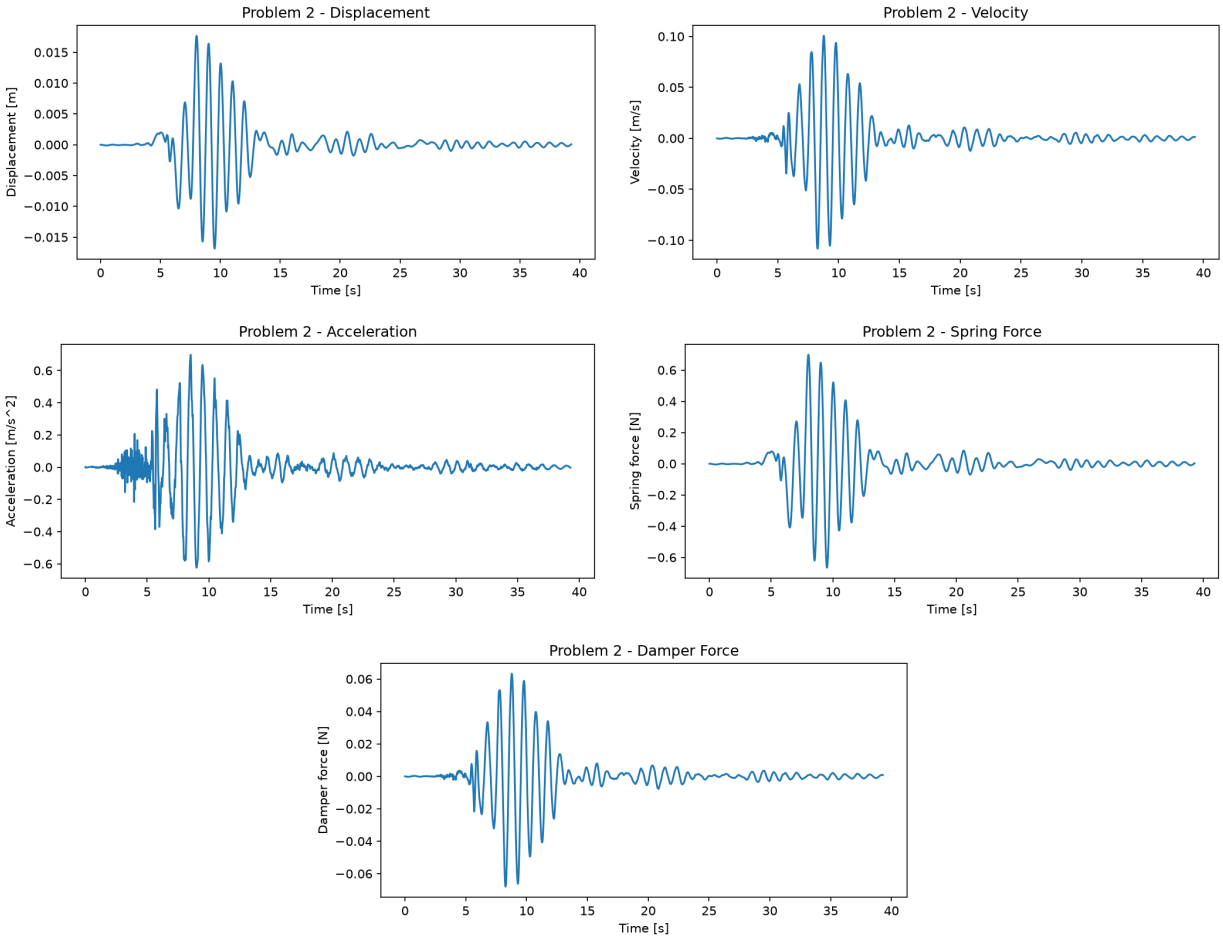


Figure 3: Problem 2: time histories.

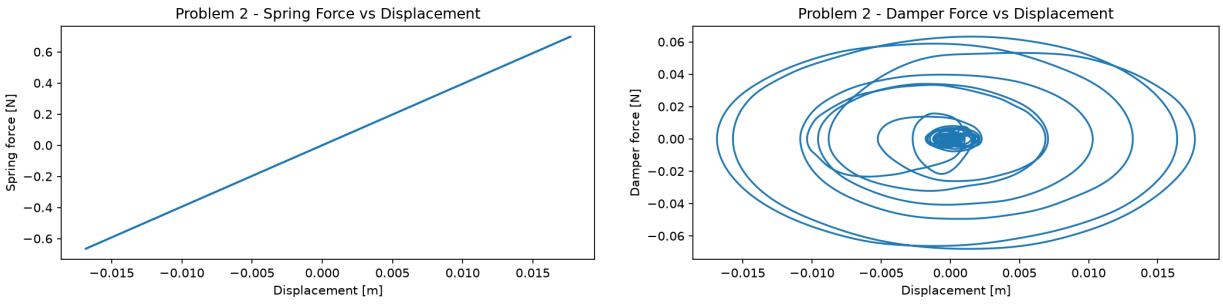


Figure 4: Problem 2: displacement vs spring force and damper force.